



HSP: A Hierarchical Semantic Preserving method for multimodal entity alignment

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ABSTRACT

Multimodal entity alignment (MEA) aims to identify equivalent entities across multimodal knowledge graphs. However, existing entailment-based approaches fail to capture entity-aware cross-modal semantics due to coarse representation and rigid fusion. In this paper, we propose an innovative Hierarchical Semantic-Preserving (HSP) method, which holistically addresses semantic collapse, cross-modal misalignment, and rigid fusion by preserving high-dimensional visual structure, calibrating heterogeneous representations in a shared semantic space, and adapting fusion dynamically to entity types. Specifically, HSP first designs a hierarchical vision-preserving encoder to preserve the intrinsic topology of high-dimensional visual representations, ensuring structural integrity and fine-grained semantic retention in the preserve stage. Then, HSP constructs a cross-modal semantic calibration network to align textual and visual modalities within a unified semantic space in the calibrate stage, mitigating cross-modal discrepancies. Finally, in the adapt stage, HSP develops an entity-type-adaptive fusion controller to dynamically balance modality contributions through meta-learning, enabling flexible, type-aware multimodal integration. Extensive experiments conducted on three public cross-lingual MEA benchmark datasets (ZH-EN, JA-EN, FR-EN) and two cross-graph datasets (FB-DB, FB-YG) under three experimental settings demonstrate that the proposed model achieves up to a 2% improvement in MRR over strong baseline methods.

1. Introduction

Knowledge graphs (KGs) provide a structured representation of semantic knowledge by mapping real-world entities to graph nodes, with edges explicitly modeling the relationships among them. Since the deployment of large-scale knowledge graphs [1], numerous studies have advanced their construction and application, particularly in tasks such as question answering [2], recommender systems [3], and natural language understanding [4], where KGs have become indispensable tools. However, with the rapid proliferation of multimedia content on social media platforms, traditional text-based knowledge graphs exhibit inherent limitations in leveraging and representing multimodal information. To address this issue, Chen et al. [5] propose Multimodal Knowledge Graphs (MMKGs) associating entities and relations with multimodal signals, enabling effective representation of real-world scenarios through cross-modal information aggregation. MMKGs enable richer and more context-aware knowledge representations, thereby supporting advanced applications such as visual question answering [6,7]

and embodied intelligence [8,9]. At the same time, they introduce new challenges for existing research due to the complexity of their construction. In particular, the multimodal entity alignment (MEA) task has become a focal point, since effective alignment of entities across knowledge graphs significantly enhances semantic interoperability, as illustrated in Fig. 1.

To tackle this complex task, Chen et al. [5] initially propose the classical three-stage architecture consisting of embedding, fusion, and matching, followed by subsequent studies [10–12] that develop various methods to exploit visual clues from the image modality and integrate multimodal information, partially alleviating the scarcity of seed alignments. More recent works [11,13,14] further concentrate on the utilization of fused multimodal representations and the design of more sophisticated fusion strategies. Nonetheless, Zhao et al. [15] reveal that these approaches still suffer from insufficient inter-KG interaction and are vulnerable to multimodal heterogeneity. To fundamentally address these limitations at a paradigm level, they reframe the multimodal

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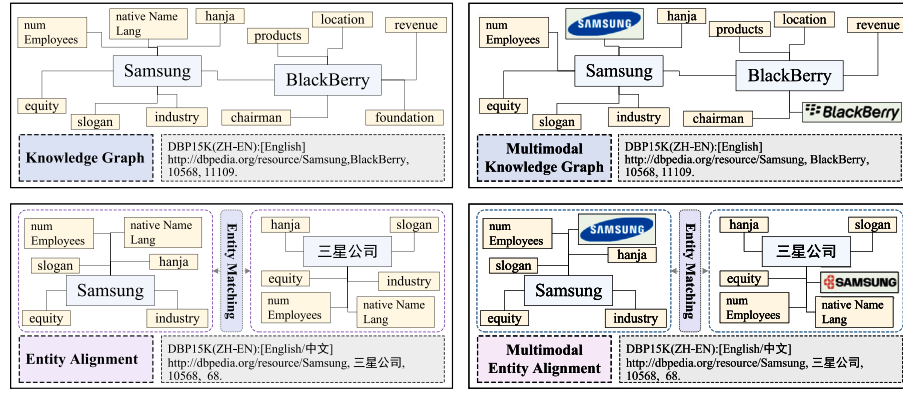


Fig. 1. Detailed presentation of different knowledge graph and entity alignment tasks.

entity alignment task as an entailment reasoning problem, where the multimodal description of one entity semantically entails that of another, implying that they correspond to the same real-world entity. By projecting information from all modalities into the textual space of a pre-trained language model, this entailment-based formulation enables fine-grained, token-level interactions between entity representations. Consequently, it achieves more robust cross-modal semantic alignment and demonstrates significantly superior performance compared to contemporary methods [16–18].

However, the aforementioned entailment-based framework inevitably encounters three critical deficiencies: (1) dimensionality catastrophe in cross-modal semantic mapping, (2) lack of cross-modal semantic consistency, and (3) limited adaptability in modality-specific fusion. Specifically, visual modalities encode rich fine-grained semantics, such as spatial layout, facial details, or architectural cues, but existing entailment-based methods typically compress these high-dimensional features into low-dimensional textual spaces through linear projection, resulting in substantial semantic loss. Meanwhile, visual representations, which are continuous, contrast sharply with the discrete, symbolic form of textual descriptions, and without a calibrated semantic bridge, models often align entities based on spurious co-occurrence rather than genuine semantic correspondence. Moreover, current fusion strategies apply uniform weighting across all entity types, ignoring that “person” entities rely heavily on facial details, “building” entities benefit from global structural patterns, and “abstract concept” entities may lack meaningful visual signals, rendering the fusion process rigid and suboptimal.

To address these limitations and advance the entailment paradigm, we propose a novel **Hierarchical Semantic Preserving (HSP)** method that reconstructs the alignment process via a three-stage “Preserve–Calibrate–Adapt” mechanism. **Specifically**, HSP comprises three key components. First, the Hierarchical Vision-Preserving Encoder (HVPE) preserves the intrinsic topology of 2048-dimensional visual features through a residual hierarchical Transformer with three parallel paths capturing local, global, and semantic information. Second, the Cross-Modal Semantic Calibration Network (CSCN) projects visual and textual representations into a shared semantic subspace via bilinear mapping and contrastive learning with InfoNCE loss and hard negative mining. Finally, the Entity-Type Adaptive Fusion Controller (EAFC) leverages meta-learning to generate entity-type-aware fusion weights via a gated attention mechanism, enabling context-sensitive, type-aware fusion across modalities. These components form a unified, interdependent pipeline, where high-fidelity features support cross-modal calibration, and calibrated representations are refined through type-aware fusion, forming a coherent end-to-end framework for multimodal entity alignment.

To comprehensively evaluate the proposed model and its hierarchical semantic preservation (HSP) strategy, we conduct systematic experiments on three cross-lingual datasets (ZH–EN, JA–EN, FR–EN)

and two cross-graph datasets (FB15K–DB15K, FB15K–YG15K) under three seed settings (20%, 50%, 80%), where results show that HSP achieves up to 2.0% MRR improvements over the strongest baselines on cross-graph benchmarks under low-resource settings, while yielding consistent gains of 0.2%–0.6% on cross-lingual datasets, demonstrating robust generalization across diverse language pairs. The remainder of this paper is organized as follows: Section 2 reviews related work on multimodal knowledge graphs and entity alignment. Section 3 presents the proposed model, detailing the HVPE, CSCN, and EAFC modules. Section 4 describes the experimental setup and reports empirical results, including ablation studies. Section 5 discusses limitations and outlines directions for future work. Finally, Section 6 concludes this work.

In summary, the main contributions of this paper are as follows:

- We identify three key limitations in existing entailment-based approaches that hinder effective cross-modal alignment across modalities: dimensionality imbalance, semantic inconsistency, and inflexible modality fusion.
- We propose Hierarchical Semantic-Preserving (HSP), an entailment-based multimodal entity alignment method that introduces a three-stage “Preserve–Calibrate–Adapt” paradigm to jointly mitigate dimensionality-induced semantic collapse; enforce cross-modal semantic consistency; and enable entity-type-aware adaptive fusion.
- Extensive experiments on multiple public MEA benchmarks demonstrate that HSP achieves superior and consistently robust performance across both cross-lingual and cross-graph alignment scenarios.

2. Related works

In this section, we systematically review the developmental trajectory of multimodal knowledge graphs, and then elaborate on the specifics of the multimodal entity alignment task.

2.1. Multimodal knowledge graph

Multimodal knowledge graphs integrate heterogeneous modalities such as text, images, audio, and video, overcoming the semantic limitations of traditional text-based knowledge graphs [19]. Such graphs enable fine-grained, multi-perspective semantic modeling of entities while enhancing semantic richness and cross-modal understanding. Recent studies have explored this paradigm from multiple perspectives. Zhang et al. [20] propose an optimal transport-based knowledge graph embedding model to address a key challenge in multimodal knowledge graph embedding: direct fusion of heterogeneous modalities can disrupt their intrinsic spatial structures, hindering semantic consistency

and comprehensive representation. Zhao et al. [21] develop a multi-modal knowledge graph-based image captioning framework to mitigate weak entit-visual associations caused by long-tail named entities and complex visual contexts. Meanwhile, Chen et al. [22] introduce a transformer-based architecture with a modality-level noise masking mechanism for robust multimodal entity feature fusion.

2.2. Multimodal entity alignment

Multimodal Entity Alignment (MEA) extends the traditional entity alignment task from knowledge graphs to multimodal knowledge graphs by integrating heterogeneous modalities, such as text and images, thereby enabling cross-graph multimodal entity matching. Early work [5] introduces a three-stage paradigm comprising embedding encoding, feature fusion, and feature matching, establishing a foundational framework for subsequent research. Later studies [11,23] recognize the impact of cross-modal joint embeddings on alignment accuracy. EVA [24] leverages visual information as anchors to perform unsupervised entity alignment, enhancing matching performance. MCLEA [23] applies contrastive learning strategies to efficiently learn multimodal entity representations, further improving alignment quality. IBMEA [25] incorporates a variational information bottleneck to reduce interference from redundant information and misleading cues within the multimodal fusion process. More recent research [26,27] continues to explore advanced modality fusion strategies: the former dynamically computes modality weights and calibrates them according to uncertainty, whereas the latter addresses weak visual signals and modality imbalance through cross-modal mapping combined with joint optimization losses.

In summary, existing approaches have achieved substantial advances in modality encoding and cross-modal fusion, yet they remain insufficient for capturing fine-grained cross-modal interactions. The proposed method addresses inherent limitations of the entity-centric framework, enabling simultaneous enhancement of fine-grained cross-modal semantic consistency and alignment accuracy.

3. Methodology

3.1. Task objective

The objective of the proposed HSP model is to address three key limitations in existing entailment-based multimodal entity alignment methods: (1) semantic distortion caused by excessive compression of high-dimensional visual features into a text-centric space, (2) insufficient cross-modal interaction leading to weak semantic alignment, and (3) static fusion that ignores entity-type-specific modality relevance. By mitigating these issues, HSP better preserves feature fidelity, cross-modal consistency, and entity-type awareness, thereby enhancing visual-language co-modeling performance.

3.2. Overview

The Hierarchical Semantic-Preserving (HSP) model addresses key limitations in existing multimodal entity alignment approaches through a *Preserve-Calibrate-Adapt* architecture that organizes feature processing into a progressive and interdependent pipeline. As illustrated in Fig. 2, HSP first preserves the hierarchical structure of visual features in their original high-dimensional space to mitigate information loss, thereby providing high-fidelity inputs for cross-modal alignment. It then aligns heterogeneous visual and textual modalities into a shared semantic subspace via contrastive calibration, establishing a consistent semantic basis. Finally, it adaptively fuses multimodal representations according to entity types to produce unified embeddings. These embeddings are subsequently fed into a pre-trained language model for entailment classification, which outputs a confidence score to guide the final alignment decision.

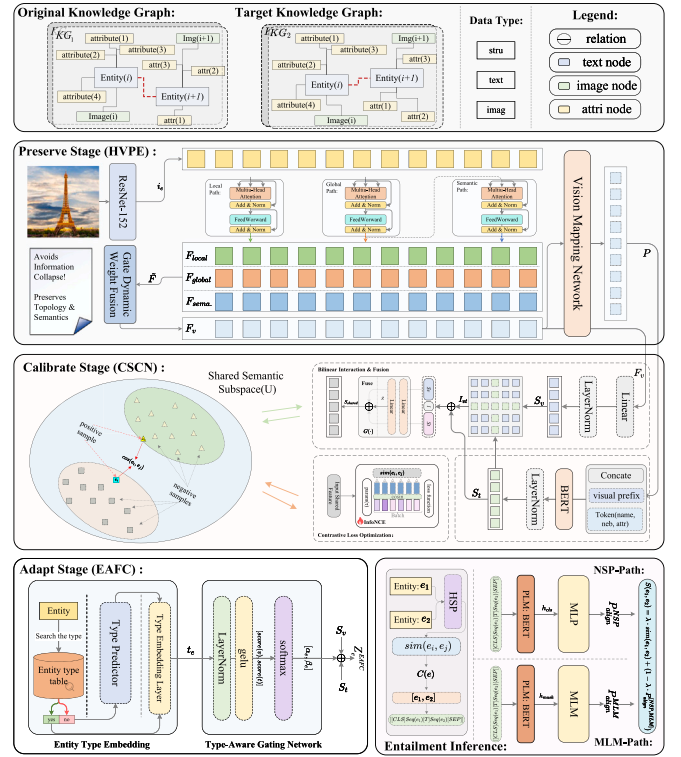


Fig. 2. The overall architecture of the proposed HSP model. Different rounded rectangular areas contain different components to demonstrate their working principles. The lower right corner area is the process of inference. The top area is the input data for the model.

3.3. Hierarchical vision-preserving encoder

To mitigate the dimensional catastrophe and semantic collapse in the prior method [15], which naively compress high-dimensional visual features into low-dimensional spaces, we design the Hierarchical Vision-Preserving Encoder (HVPE). Unlike prior approaches that directly project visual features such as the ResNet-152 outputs into low-dimensional PLM embedding spaces, HVPE adopts a two-stage architecture: the first stage operates in the original high-dimensional feature space to preserve topological structure and semantic richness, while the second stage generates PLM-compatible learnable prefixes to enable effective downstream fusion and reasoning.

3.3.1. Hierarchical feature enhancement

In the first stage, HVPE processes the feature vector i_e , extracted from the raw image v_e of entity e using a frozen ResNet-152 network.

$$F^0 = \text{ResNet152}(v_e), \quad F^0 \in \mathbb{R}^{H \times W \times d_v} \quad (1)$$

$$i_e = \text{Flatten}(F^0), \quad i_e \in \mathbb{R}^{N \times d_v} \quad (2)$$

where $d_v = 2048$ denotes the dimensionality of the flattened feature vector, and $N = H \times W$ represents the spatial resolution of the feature map before flattening.

Then, we design a residual hierarchical Transformer with three pathways to generate enriched high-dimensional features while preserving the image topology. The local pathway F_{local} employs window-based attention to capture fine-grained visual patterns such as textures and edges. The global pathway F_{global} utilizes full self-attention to model overall composition and spatial relationships across the image. The semantic pathway F_{semantic} introduces a lightweight multilayer

perceptron for semantic aggregation, extracting high-level semantic concepts such as “person” or “building”.

$$F_{\text{local}} = \text{transformer}_{\text{local}}(i_e) \quad (3)$$

$$F_{\text{global}} = \text{transformer}_{\text{global}}(i_e) \quad (4)$$

$$F_{\text{semantic}} = \text{transformer}_{\text{semantic}}(F_{\text{global}}) \quad (5)$$

To ensure the integrity of information flow and the effective utilization of multi-granularity features, we introduce cross-layer feature reuse to integrate the outputs F_{local} , F_{global} , and F_{semantic} . Subsequently, a learnable dynamic weighting gating network adaptively generates a normalized weight vector based on the input image content, and performs weighted fusion over the three outputs to obtain the final high-dimensional visual representation.

$$\tilde{F} = G(\text{Pool}(F_{\text{local}}), \text{Pool}(F_{\text{global}}), \text{Pool}(F_{\text{semantic}})) \in \mathbb{R}^{d_c} \quad (6)$$

$$w_p = \text{softmax}(W_p \cdot \text{LN}(\tilde{F}) + b_p) \quad (7)$$

$$F_v = \sum_{p \in \{\text{loc}, \text{glo}, \text{sem}\}} w_p \cdot \text{Pool}(\tilde{F}) \quad (8)$$

where $\text{Pool}(\cdot)$ is the learnable attention pooling operation, and $G(\cdot)$ is the learnable gate network.

3.3.2. PLM-compatible visual prefix generation

In the second stage, to facilitate cross-modal interaction within the PLM while preserving high-dimensional information, we introduce a learnable low-dimensional prefix sequence instead of directly compressing the full visual features. Specifically, we design a vision mapping network (VMN), implemented as a multi-layer perceptron, to project the extracted visual features F_v into a learnable token embedding sequence of length L :

$$P = \text{VMN}(F_v) \in \mathbb{R}^{L \times d}, \quad P = [P_1, P_2, \dots, P_L]. \quad (9)$$

The resulting tokens are prepended to the textual sequence of each entity and serve as input embeddings for the PLM. Unlike prior approaches that rely on aggressive compression, HVPE preserves the original visual representations without discarding high-dimensional information.

3.4. Cross-modal semantic calibration network

To mitigate the modality discrepancy between continuous visual representations and discrete textual semantics, we propose the Cross-Modal Semantic Calibration Network (CSCN). Its core objective is to project the high-dimensional visual representations produced by HVPE and the PLM-encoded contextual text representations T_e into a shared semantic subspace. Semantic correspondence is further strengthened through contrastive learning. Unlike token-level interactions within the PLM [15], CSCN explicitly models and corrects cross-modal biases through calibration and fusion modules, yielding a robust shared semantic representation. The process consists of three steps: textual context modeling, cross-modal semantic mapping and calibration, and cross-modal contrastive learning.

3.4.1. Contextual text representation

For a given entity e , its textual information includes the name sequence, neighbor sequence, and attribute sequence. These sequences are processed by a tokenizer to obtain a set of tokens. Meanwhile, the learnable prefix token sequence P generated by HVPE is inserted at the beginning of the text sequence, thereby constructing the input sequence for the PLM:

$$X_e = [p_1, \dots, p_L, t_1, \dots, t_m] \in \mathbb{R}^{(L+m) \times d} \quad (10)$$

where d denotes the embedding dimension of the PLM, and m is the number of text tokens. After passing through L layers of the Transformer encoder, this sequence X_e produces a contextualized hidden state matrix:

$$H_t = \text{PLM}(X_e) = [h_1, h_2, \dots, h_{L+m}] \in \mathbb{R}^{(L+m) \times d} \quad (11)$$

To obtain the overall textual representation of the entity, we apply attention pooling over H_t :

$$u_i = q^\top \cdot \tanh(W \cdot h_i + b) \quad (12)$$

$$T_e = \sum_{i=1}^{L+m} \frac{\exp(u_i)}{\sum_{j=1}^{L+m} \exp(u_j)} \cdot h_i \quad (13)$$

where q is a learnable query vector, W denotes the projection parameters, h_i is the representation of the i th token in H_t , and T_e corresponds to the contextual text representation of entity e .

3.4.2. Cross-modal semantic projection and calibration

After obtaining the visual aggregated representation from HVPE and the textual representation T_e , CSCN first projects these two features into a shared semantic subspace of dimension d_c :

$$S^v = \text{LayerNorm}(\sigma(W_v^\top \cdot F_v) + b_v) \quad (14)$$

$$S^t = \text{LayerNorm}(\sigma(W_t^\top \cdot T_e) + b_t) \quad (15)$$

where $W_v \in \mathbb{R}^{d_c \times d_c}$, $W_t \in \mathbb{R}^{d_c \times d_c}$ are the learnable weight matrices, and σ is a nonlinear activation function.

To further enhance fine-grained cross-modal semantic coupling, we introduce a bilinear interaction term to ensure the independent semantics of visual and textual features are preserved, while also achieving cross-modal alignment. The final cross-modal shared representation $S_{\text{shared}} \in \mathbb{R}^{d_c}$ is obtained through a gating mechanism as follows:

$$I_{vt} = (S^v)^\top \cdot U \cdot S^t \quad (16)$$

$$g = \sigma(W_g \cdot [S^v; S^t; I_{vt}] + b_g) \quad (17)$$

$$S_{\text{shared}} = g \cdot S^v + (1 - g) \cdot S^t \quad (18)$$

where $U \in \mathbb{R}^{d_c \times d_c}$ is a learnable interaction matrix, and $W_g \in \mathbb{R}^{1 \times (2d_c + 1)}$ and b_g are learnable parameters.

3.4.3. Contrastive calibration and fusion

In the third stage, we employ a contrastive learning mechanism to explicitly enhance the consistency between visual and textual representations in the shared subspace. For an entity e in a batch, its positive sample pair is (S_e^v, S_e^t) , and the negative sample set consists of the representations of other entities in the same batch. The InfoNCE loss is defined as follows:

$$\mathcal{L}_{\text{cscn}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(\hat{S}_i^v, \hat{S}_i^t)/\tau)}{\sum_{j=1}^B \exp(\text{sim}(\hat{S}_i^v, \hat{S}_j^t)/\tau)} \quad (19)$$

where $\text{sim}(\cdot, \cdot)$ denotes cosine similarity, and τ is the temperature coefficient.

After completing the contrastive calibration, the calibrated visual and textual representations S^v and S^t are passed to the EAFC module for type-aware adaptive fusion.

3.5. Entity-type adaptive fusion controller

The HVPE and CSCN modules preserve high-dimensional features and perform cross-modal semantic calibration but do not explicitly model heterogeneous modality dependencies across entity types. For instance, visual signals (e.g., facial images) are often more discriminative

for Person entities, whereas textual signals (e.g., attribute descriptions or neighbor relations) convey stronger semantics for Organization and Location entities. Static fusion strategies [15] overlook such type-specific dependencies, increasing the risk of semantic misalignment. Therefore, a type-aware fusion mechanism is essential for robust cross-modal alignment.

Therefore, we propose the Entity-Type Adaptive Fusion Controller (EAFC). Built upon the calibrated visual representation S^v and textual representation S^t produced by CSCN, EAFC introduces a type-aware dynamic weighting mechanism that adaptively adjusts the fusion ratio according to the entity type t_e . In addition, to improve the model's generalization capability on few-shot and unseen entity types, EAFC employs a meta-learning-based adaptive optimization strategy.

3.5.1. Type-aware fusion weight generation

In the first stage, EAFC leverages the entity type embedding to generate adaptive weights for different modalities. If the type t_e of entity e is known, the corresponding type embedding vector is retrieved from a lookup table:

$$\mathbf{t}_e = \text{Embed}(t_e) \in \mathbb{R}^{d_t} \quad (20)$$

where d_t denotes the dimensionality of the type embedding. If the entity type is unknown, a lightweight type predictor (e.g., a classifier based on the PLM [CLS] vector) is used to estimate its type distribution $p(t | e)$, and the weighted average embedding is then computed as:

$$\mathbf{t}_e = \sum_{k=1}^K p(t_k | e) \cdot \text{Embed}(t_k) \quad (21)$$

where K denotes the total number of predefined entity types. The type embedding $\mathbf{t}_e$ is then projected through a gating network $G(\cdot)$ to generate normalized fusion weights for the visual and textual modalities.

$$[\alpha_e, \beta_e] = \text{softmax}(G(\mathbf{t}_e)) \in \mathbb{R}^2, \quad (\alpha_e + \beta_e = 1) \quad (22)$$

$$Z_e^{\text{EAFC}} = \alpha_e \cdot S^v + \beta_e \cdot S^t \quad (23)$$

3.5.2. Meta-learning based adaptive optimization

In the second stage, we further improve the generalization capability of EAFC toward few-shot and previously unseen entity types by integrating a MAML-based meta-optimization strategy. This strategy equips EAFC with a transferable initialization, enabling rapid specialization to emerging entity types under limited supervision. Specifically, each entity type k is formulated as an independent meta-task T_k , aiming to learn a fusion weight generator $G(\cdot)$ that can adapt effectively from scarce samples. The training procedure follows a bi-level optimization paradigm consisting of an inner loop and an outer loop.

Inner Loop (Task-Specific Adaptation): For each task T_k , a small support set $\mathcal{D}_k^{\text{sup}}$ is sampled to emulate the low-resource setting. Initialized with the global parameters θ , EAFC performs a gradient-based update to rapidly specialize the fusion behavior, yielding task-adapted parameters θ'_k :

$$\theta'_k = \theta - \alpha \nabla_{\theta} \mathcal{L}_{T_k}(\theta; \mathcal{D}_k^{\text{sup}}), \quad (24)$$

Outer Loop (Meta-Level Optimization): The task-adapted parameters are evaluated on the query set $\mathcal{D}_k^{\text{qry}}$ of task T_k , encouraging the model to favor parameter initializations that generalize after adaptation. The global parameters θ are then optimized to minimize the query loss:

$$\theta \leftarrow \theta - \beta \nabla_{\theta} \mathcal{L}_{T_k}(\theta'_k; \mathcal{D}_k^{\text{qry}}). \quad (25)$$

where α and β denote the learning rates for the inner loop and outer loop, respectively, and $\mathcal{L}_{T_k}$ represents the loss function for task T_k .

3.6. Unified input construction and training

Through the integration of HVPE, CSCN, and EAFC, we obtain a robust multimodal entity representation Z_e^{EAFC} for each entity e .

However, representation learning alone is insufficient for the downstream entity alignment task. To facilitate effective alignment learning, we introduce a unified input sequence construction mechanism coupled with a multi-task joint optimization objective, leveraging the PLM to enable end-to-end training.

3.6.1. Multimodal input sequence construction

For each entity e , we construct an input sequence that includes visual prefixes and textual information (name, neighbors, and attributes):

$$\text{Seq}(e) = [P_e; \text{Token}(S_{\text{name}}); \text{Token}(S_{\text{nbr}}); \text{Token}(S_{\text{attr}})] \quad (26)$$

where P denotes the visual prefix tokens generated by HVPE, and the textual part is tokenized into a sequence of tokens.

In the candidate alignment task, for a source entity e and its candidate target entity e' , we construct the input template as follows:

$$T(e, e') = [\text{CLS}] \text{Seq}(e) [\text{T}] \text{Seq}(e') [\text{SEP}] \quad (27)$$

where $[\text{T}]$ serves as a delimiter to ensure the distinguish ability of sequences across entities. The sequence is then fed into the PLM for encoding, producing the contextual hidden state matrix.

3.6.2. Triple construction

To enhance discriminative cross-modal entity alignment, we formulate a triplet-based objective within a multi-task joint optimization framework. For a source entity e , its aligned counterpart e^+ is paired with multiple negative samples e^- . The negatives are constructed via a hard sampling strategy that combines random negatives with highly similar yet unaligned candidates in the embedding space. This design promotes stable optimization and improves fine-grained discriminative learning among semantically similar entities.

3.6.3. Joint optimization objectives

After obtaining the triplets, we design a joint optimization loss function at multiple levels:

- The Embedding Alignment Loss is used to constrain positive sample pairs to be closer in the embedding space and is defined as follows:

$$\mathcal{L}_{\text{mr}} = \sum_{(e, e^-, e^+)} \max(0, d(Z_e^{\text{EAFC}}, Z_{e^+}^{\text{EAFC}}) - d(Z_e^{\text{EAFC}}, Z_{e^-}^{\text{EAFC}}) + m) \quad (28)$$

where $d(\cdot, \cdot)$ denotes the squared Euclidean distance.

- The Entailment Losses are used to discriminate and rank alignment relations and are defined as follows:

$$\mathcal{L}_{\text{be}} = -\left(y \cdot \log p(y | T(e, e^+)) + (1 - y) \cdot \log(1 - p(y | T(e, e^-)))\right) \quad (29)$$

$$\mathcal{L}_{\text{bm}} = \sum_{(e, e^-, e^+)} \max(0, p(y | T(e, e^-)) - p(y | T(e, e^+)) + m') \quad (30)$$

- The Cross-Modal Contrastive Loss is employed to ensure consistency between visual and textual representations in the shared subspace and is defined as follows:

$$\mathcal{L}_{\text{cscn}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\text{sim}(\hat{S}_i^v, \hat{S}_i^t)/\tau)}{\sum_{j=1}^B \exp(\text{sim}(\hat{S}_i^v, \hat{S}_j^t)/\tau)} \quad (31)$$

- The EAFC employs a Meta-Learning Loss to enhance generalization across different entity types and is defined as follows:

$$\mathcal{L}_{\text{meta}} = \sum_{k \in \mathcal{T}} \mathcal{L}_{T_k}(\theta'_k; \mathcal{D}_k^{\text{qry}}) \quad (32)$$

During training, we employ joint training to optimize the overall model, so the final objective function is given by:

$$\mathcal{L} = \lambda_1 \cdot \mathcal{L}_{\text{mr}} + \lambda_2 \cdot \mathcal{L}_{\text{be}} + \lambda_3 \cdot \mathcal{L}_{\text{bm}} + \lambda_4 \cdot \mathcal{L}_{\text{cscn}} + \lambda_5 \cdot \mathcal{L}_{\text{meta}} \quad (33)$$

Table 1
Detailed statistics of datasets.

Dataset type	Dataset	MMKGs	#Ent.	#Rel.	#Attr.	#Rel. Triplet	#Attr. Triplet	#Img.	#Seed
Cross-graph	FB15K-DB15K	FB15K	14,951	1345	116	592,313	29,395	13,444	12,849
		DB15K	12,842	279	225	89,197	48,080	12,837	
	FB15K-YG15K	FB15K	14,951	1345	116	592,313	29,395	13,444	11,199
		YG15K	15,404	32	7	122,886	23,532	11,194	
Cross-lingual	DBP _{ZH-EN}	Chinese(ZH)	19,388	1701	7780	70,414	379,684	19,388	15,000
		English(EN)	19,572	1323	6933	95,142	567,755	19,572	
	DBP _{JA-EN}	Japanese(JA)	19,814	1299	5681	77,214	354,619	12,739	15,000
		English(EN)	19,780	1153	5850	93,484	497,230	13,741	
	DBP _{FR-EN}	French(FR)	19,661	903	4431	105,998	528,665	14,174	15,000
		English(EN)	19,993	1208	6161	115,722	576,543	13,858	

3.7. Dual entailment reasoning

During inference, the model identifies the aligned entity in the target knowledge graph G_2 for each source entity e in G_1 through a two-stage process: embedding-based candidate retrieval followed by entailment-based semantic reasoning. In the first stage, the source and target entities are encoded and fused via HVPE, CSCN, and EAFC to obtain unified cross-modal representations. Cosine similarity is then computed in the shared embedding space, and the top- k most similar entities are selected to form the candidate set $C(e)$:

$$Sim_{emb}(e, e') = \frac{Z_e^{EAFC} \cdot Z_{e'}^{EAFC}}{\|Z_e^{EAFC}\| \cdot \|Z_{e'}^{EAFC}\|}, \quad (34)$$

Although embedding similarity provides an efficient coarse-grained estimate of entity compatibility, it can become unreliable when candidates exhibit highly similar multimodal representations. To address this, HSP employs a PLM-based entailment reasoning mechanism for deeper semantic verification, constructing a joint input sequence $X(e, e')$ for each candidate $e' \in C(e)$ and computing the alignment probability using one of two complementary reasoning variants: HSP-N, which emphasizes global semantic compatibility between entity pairs, and HSP-M, which enables finer-grained contextual inference.

HSP-N (NSP-based reasoning). This variant models entity alignment as a sentence-pair task, processing the joint sequence through the PLM and using the [CLS] representation h_{CLS} to encode the global semantic relationship between the two entities, with the alignment probability then computed as:

$$p_{align}^{NSP}(e, e') = \sigma(\mathbf{w}_{NSP}^\top h_{CLS} + b_{NSP}), \quad (35)$$

HSP-M (MLM-based reasoning). This variant casts alignment as a masked semantic inference task, inserting a [MASK] token into the input sequence and predicting whether it should be filled with “Yes”, thereby enabling finer-grained semantic inference conditioned on the full multimodal context, with h_{mask} denoting the hidden representation at the masked position:

$$p_{align}^{MLM}(e, e') = \sigma(\mathbf{w}_{MLM}^\top h_{mask} + b_{MLM}), \quad (36)$$

Both variants share the same architecture and differ only in the entailment prediction head, with $p_{entail}(e, e')$ denoting either p_{align}^{NSP} or p_{align}^{MLM} depending on the reasoning variant; the final matching score then combines embedding similarity with the entailment signal:

$$S(e, e') = \gamma \cdot Sim_{emb}(e, e') + (1 - \gamma) \cdot p_{entail}(e, e'), \quad (37)$$

$$e^* = \arg \max_{e' \in C(e)} S(e, e'). \quad (38)$$

where γ balances structural similarity and semantic verification. By integrating embedding-based retrieval with entailment reasoning, HSP performs multi-level semantic validation, reducing reliance on embedding similarity alone and enhancing robustness when aligning noisy, heterogeneous, or semantically overlapping multimodal entities.

4. Experiments

4.1. Datasets

To comprehensively evaluate the performance of the proposed HSP model,¹ we conduct experiments on five representative benchmark datasets encompassing both cross-graph and cross-lingual scenarios. As summarized in Table 1, the FB15K-DB15K dataset is constructed by aligning entities between FB15K [28] and DB15K [29], resulting in a cross-kg entity alignment benchmark with 12,849 alignment seeds and 26,281 associated images. Similarly, the FB15K-YG15K dataset is constructed by matching identical entities between FB15K and YG15K [30], yielding a dataset with 11,199 alignment seeds and 24,638 associated images. The DBP dataset [29] comprises three cross-lingual sub-datasets: ZH-EN, JA-EN, and FR-EN, covering Chinese, English, French, and Japanese, and representing complex cross-lingual interaction scenarios; these sub-datasets contain 38,960, 26,480, and 28,032 associated images, respectively, providing rich cross-modal interaction cues for entity alignment. For training, we adopt three different alignment seed ratios: 20%, 50%, and 80%, and for entities without associated images, we follow the prior work [24] setting by assigning random vectors as their visual features.

4.2. Implementation details

All experiments are implemented using the PyTorch framework² and conducted on a system equipped with two NVIDIA RTX 4090 GPUs, each with 24 GB of memory. To ensure reproducibility, we repeat each experiment using five distinct random seeds and report the average performance across three independent runs for each seed. The model is trained with a batch size of 32 to balance training stability and GPU memory consumption, for up to 100 epochs, with early stopping applied based on validation set performance to mitigate overfitting. We employ the Adam optimizer, using an initial learning rate of 3×10^{-5} for the language encoder and 1×10^{-4} for other components; this learning rate is decayed by a factor of 0.1 every 20 epochs. A dropout rate of 0.1 and weight decay of 1×10^{-5} are applied throughout the model to enhance generalization, and the GeLU activation function is used in all applicable layers.

To ensure a fair comparison with the state-of-the-art model [15], we reproduce its results using the original protocol. The original work used NVIDIA A6000 GPUs with 48 GB memory,³ while our experiments use RTX 4090 GPUs. Differences in hardware may cause minor performance variations. We retain the original results for reference and analyze discrepancies to support a transparent evaluation of HSP.

¹ <https://github.com/SheetahKing/HSP>.

² <https://pytorch.org/>.

³ <https://github.com/OreOZhao/ME3A>.

4.3. Evaluation metrics

To ensure a comprehensive evaluation of the proposed method on the multimodal entity alignment task, we adopt Hits@n and MRR as primary evaluation metrics. Hits@n measures the proportion of test entities for which the correct alignment appears within the top n ranked candidates; for instance, Hits@1 = 0.8 indicates that the correct match is ranked first in 80% of the cases. MRR, or Mean Reciprocal Rank, computes the average of the reciprocal ranks of correct entities, assigning scores of 1, 0.5, 0.33, and so on for ranks 1, 2, 3, respectively. This metric rewards models that place correct matches higher in the ranked list. Higher values of both Hits@n and MRR indicate more accurate and robust entity alignment performance.

4.4. Baseline models

To evaluate the effectiveness of the proposed model, we assess it against several recent and representative baselines on two cross-graph datasets. All hyperparameters follow the configurations reported in the original papers to ensure reproducibility. These baselines cover diverse paradigms, including multimodal fusion, contrastive learning, distribution alignment, and causal modeling: PoE [31] establishes a foundational framework for cross-modal semantic fusion; MMEA [5] unifies structure, attributes, and visual information for multimodal entity alignment; EVA [24] employs vision as a pivot for unsupervised alignment; MCLEA [23] advances multimodal contrastive learning to align heterogeneous embedding spaces; MEAformer [32] formulates a meta-modality hybrid mechanism for heterogeneous modality fusion; PCMEA [18] introduces pseudo-label calibration to alleviate noise propagation; LoginMEA [33] presents a local-to-global interaction network for multi-level relational fusion; OTMEA [16] aligns multimodal distributions via optimal transport to reduce heterogeneity; GSIEA [34] separately models graph structure and multimodal information to mitigate structural heterogeneity; RICEA [26] enhances generalization through relative interaction modeling and dynamic calibration; CDMEA [17] addresses modality bias from a causal perspective; ME³A [15] formulates MEA as an entity entailment task to enable fine-grained cross-graph interaction. On the DBP dataset, we evaluate the model against MSNEA [35], MultiJAF [36], DESAlign [37], PSNEA [38], AMFSEA [13], and IBMEA [25].

4.5. Main results

We conduct extensive experiments across multiple datasets to evaluate the performance of the proposed HSP model against recent and competitive baselines. As shown in Table 2, HSP demonstrates highly competitive performance on the cross-graph datasets FB15K-DB15K and FB15K-YG15K under alignment seed ratios of 20%, 50%, and 80%, frequently surpassing previous methods. In the low-resource setting with only 20% seed alignments, although some methods achieve slightly higher Hits@1 on FB15K-DB15K, HSP-M attains the best Hits@10 (0.837) and MRR (0.702), indicating stronger overall ranking capability and more reliable candidate alignment under limited supervision. As supervision increases, HSP's advantage becomes more evident; for instance, with 50% seeds on FB15K-DB15K, HSP improves Hits@1 to 0.755, outperforming the strongest baseline (0.735) by 2.0%, and maintains top performance across nearly all metrics at the 80% setting. These results suggest that HSP scales effectively with additional supervision while preserving stable performance gains.

According to the results in Table 3, HSP consistently achieves leading performance across all language pairs (ZH-EN, JA-EN, FR-EN). In particular, HSP-M attains an MRR of 0.946 and Hits@1 of 0.906 on FR-EN, while consistently outperforming recent competitive baselines, including the ME3A variants, on the other subsets. These uniform improvements across linguistically diverse datasets indicate that HSP enhances both the discriminability of entity representations and the accuracy of alignment, which can be attributed to its coordinated integration of structural preservation, semantic calibration, and adaptive feature fusion.

4.6. Ablation study

To assess the contribution of each component within the proposed HSP framework, we conduct systematic ablation studies on the DBP15K benchmark datasets (ZH-EN and JA-EN). As shown in Table 4, removing any core module leads to performance declines of varying magnitudes. Specifically, eliminating HVPE (Hierarchical Visual Preservation Encoder) results in the largest drop, with Hits@1 decreasing by 0.038 on ZH-EN, suggesting that preserving high-dimensional visual topology is associated with reduced semantic representation degradation. Removing CSCN, which performs cross-modal calibration through bilinear interaction and contrastive learning, yields a performance decrease of 0.032, indicating that explicit semantic alignment within a shared subspace contributes positively to model performance. Disabling EAFC (Entity-type Adaptive Fusion Controller) lowers Hits@1 by 0.023, suggesting that dynamically adjusting modality weights based on entity semantics tends to be more effective than static fusion. In addition, removing meta-learning (w/o META) or contrastive calibration (w/o CONT) causes performance drops of 0.011 and 0.028, respectively, indicating their roles in supporting cross-entity-type generalization and cross-modal consistency. Overall, these observations are consistent with the “preserve-calibrate-adapt” design principle: HVPE is associated with maintaining structural information, CSCN supports semantic alignment across modalities, and EAFC enables context-aware fusion. Collectively, these components contribute to stable performance under low-resource and cross-lingual settings.

As shown in Table 5, we systematically evaluate the impact of different pretrained visual and textual encoders on model performance and inference efficiency. For visual encoders, performance does not increase monotonically with parameter scale: VGG-16 and ViT-base contain 138M and 87M parameters, respectively, both exceeding the 60M parameters of ResNet-152, yet they exhibit slightly inferior alignment performance. Although CLIP yields an approximate 0.6% improvement in MRR, it also introduces higher inference overhead. This observation may be related to the relative sparsity of visual signals in multimodal entity alignment, where most entities are associated with only a single image, thereby limiting the effective utilization of large-scale visual models. For textual encoders, replacing the static word embeddings of GloVe with contextualized pretrained language models leads to a substantial performance gain, improving MRR by approximately 5.9%, suggesting that dynamic semantic modeling helps capture entity contextual information. In contrast, BERT-large provides only a modest improvement of about 0.4% over BERT-base, indicating that further scaling the model yields limited returns under the current data conditions. Overall, smaller encoders may be insufficient for modeling cross-modal semantics, whereas excessively large models are more likely to be constrained by data availability while incurring additional computational cost. Therefore, ResNet-152 and BERT-base offer a more balanced trade-off between performance and efficiency and are adopted as the default configuration.

4.7. Case study

To illustrate HSP's type-aware fusion, we examine two representative entity types, Person and Location (Fig. 3(a)). Unlike prior methods that rely on static fusion [15], HSP dynamically adjusts modality weights via EAFC to reflect entity-specific importance.

For *Person* entities (e.g., “Michelle Obama”), identity discrimination relies primarily on facial geometry, leading EAFC to assign greater weight to the visual modality. HVPE preserves fine-grained geometric features, such as interocular distance and jawline structure, in a 2048-dimensional space. CSCN then aligns these visual cues with bilingual textual representations in a shared semantic subspace, facilitating accurate cross-modal alignment. In contrast, for *Location* entities (e.g., “Notre-Dame de Paris”), visual appearance alone is often

Table 2

The experiment results on the DB15K-DB15K and FB15K-YG15K dataset. The best results are highlighted in bold, while the second best are underlined. The $\Delta Imp.$ represents the difference between the best and second best performances. The $\Delta Std.$ is the standard deviation of the average performance.

Seeds	Model	Ref.	FB15K-DB15K			FB15K-YG15K		
			Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
20%	PoE	ESWC'19	0.126	0.251	0.170	0.250	0.495	0.334
	MMEA	KSEM'20	0.265	0.541	0.357	0.234	0.480	0.317
	EVA	AAAI'21	0.199	0.448	0.283	0.153	0.361	0.224
	MCLEA	COLING'22	0.295	0.582	0.393	0.254	0.484	0.332
	MEAformer	MM'23	0.417	0.715	0.518	0.327	0.595	0.417
	PCMEA ^a	AAAI'24	0.436	0.707	0.528	0.397	0.654	0.486
	LoginMEA ^a	ECAI'24	0.536	0.784	0.667	0.572	0.758	0.652
	OTMEA	ICASSP'25	0.443	0.712	0.523	0.395	0.628	0.477
	GSIEA	IPM'25	0.580	0.791	0.649	0.515	0.728	0.588
	RICEA	ACL'25	0.471	0.720	0.557	0.411	0.658	0.497
	CDMEA ^a	SIGIR'25	<u>0.549</u>	0.741	0.611	0.434	0.682	0.523
	ME ³ A-N	IPM'25	0.492	<u>0.824</u>	<u>0.689</u>	<u>0.587</u>	<u>0.766</u>	0.651
	ME ³ A-M	IPM'25	0.502	0.822	0.683	0.586	0.764	<u>0.659</u>
	HSP-N	–	0.544	0.831	0.699	0.597	0.784	0.679
	HSP-M	–	0.542	0.837	0.702	0.609	0.791	0.672
	$\Delta Avg.$	–	–0.036	0.013	0.013	0.022	0.025	0.020
	$\Delta Std.$	–	± 0.012	± 0.009	± 0.010	± 0.011	± 0.007	± 0.009
50%	PoE	ESWC'19	0.464	0.658	0.533	0.411	0.669	0.498
	MMEA	KSEM'20	0.417	0.703	0.512	0.403	0.645	0.486
	EVA	AAAI'21	0.334	0.589	0.422	0.311	0.534	0.388
	MCLEA	COLING'22	0.555	0.784	0.637	0.501	0.705	0.574
	MEAformer	MM'23	0.619	0.843	0.698	0.560	0.778	0.639
	PCMEA ^a	AAAI'24	0.633	0.825	0.701	0.606	0.785	0.652
	LoginMEA ^a	ECAI'24	0.712	0.857	0.755	0.648	0.805	0.699
	OTMEA	ICASSP'25	0.658	0.854	0.734	0.617	0.809	0.687
	GSIEA	IPM'25	0.716	<u>0.886</u>	0.775	0.646	<u>0.828</u>	0.711
	RICEA	ACL'25	0.648	0.852	0.721	0.617	0.811	0.687
	CDMEA ^a	SIGIR'25	0.675	0.861	0.737	0.628	0.824	0.692
	ME ³ A-N	IPM'25	0.729	0.884	0.788	<u>0.664</u>	0.827	0.721
	ME ³ A-M	IPM'25	<u>0.735</u>	0.882	<u>0.797</u>	0.662	0.826	<u>0.726</u>
	HSP-N	–	0.747	0.893	0.804	0.674	0.837	0.736
	HSP-M	–	0.755	0.894	0.807	0.681	0.841	0.739
	$\Delta Avg.$	–	0.020	0.008	0.010	0.017	0.013	0.013
	$\Delta Std.$	–	± 0.007	± 0.005	± 0.007	± 0.008	± 0.007	± 0.006
80%	PoE	ESWC'19	0.666	0.820	0.721	0.492	0.705	0.572
	MMEA	KSEM'20	0.590	0.869	0.685	0.598	0.839	0.682
	EVA	AAAI'21	0.484	0.696	0.563	0.491	0.692	0.565
	MCLEA	COLING'22	0.735	0.890	0.790	0.667	0.824	0.722
	MEAformer	MM'23	0.765	0.916	0.820	0.703	0.873	0.766
	PCMEA ^a	AAAI'24	0.744	0.904	0.818	0.715	0.869	0.784
	LoginMEA ^a	ECAI'24	0.773	0.908	0.822	0.724	0.904	0.803
	OTMEA	ICASSP'25	0.793	0.919	0.839	0.739	0.898	0.798
	GSIEA	IPM'25	0.804	0.928	0.847	0.734	0.861	0.793
	RICEA	ACL'25	0.776	0.916	0.829	0.734	0.892	0.792
	CDMEA ^a	SIGIR'25	0.804	0.931	<u>0.859</u>	0.743	0.891	0.806
	ME ³ A-N	IPM'25	<u>0.824</u>	0.933	0.841	0.748	<u>0.935</u>	0.808
	ME ³ A-M	IPM'25	0.822	<u>0.928</u>	0.844	<u>0.751</u>	0.931	<u>0.811</u>
	HSP-N	–	0.835	0.935	0.858	0.767	0.943	0.821
	HSP-M	–	0.838	0.937	0.861	0.766	0.946	0.819
	$\Delta Imp.$	–	0.014	0.004	0.002	0.016	0.011	0.010
	$\Delta Std.$	–	± 0.006	± 0.005	± 0.004	± 0.006	± 0.004	± 0.005

^a Represents reproduced experimental results.

ambiguous, as architectural similarities among distractors such as “Milan Cathedral” and “Cologne Cathedral” may mislead vision-dominant models. HSP mitigates this by shifting emphasis to textual semantics via EAFC, prioritizing contextual descriptors such as “Paris” and “Gothic architecture”, allowing the model to disambiguate visually similar candidates and achieve robust alignment even under strong visual interference.

Additionally, we perform a quantitative analysis of modality weights across four distinct entity types. As shown in Fig. 3(b), EAFC adaptively allocates contributions of visual and textual modalities according to entity semantics. Person entities exhibit the highest visual weights, whereas Location, Organization, and Other entities display

progressively lower values, providing strong evidence for the soundness and interpretability of the proposed fusion strategy.

4.8. Efficiency analysis

Although the main experiments and ablation studies have validated HSP's effectiveness, computational efficiency remains critical for real-world deployment. We perform a component-level analysis on the large-scale OpenEA100K dataset. As shown in Table 6, the HVPE module, composed of three parallel three-layer Transformers, accounts for most of the computational overhead, while CSCN handles cross-modal projection and EAFC supports gated fusion with minimal additional

Table 3

The experiment results on the DBP15K dataset. The best results are highlighted in bold, while the second best are underlined. The $\Delta Imp.$ represents the difference between the best and second best performances. The $\Delta Std.$ is the standard deviation of the average performance.

Method	Ref.	DBP15K _{ZH-EN}			DBP15K _{JA-EN}			DBP15K _{FR-EN}		
		Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
EVA	AAAI'21	0.680	0.910	0.762	0.673	0.908	0.757	0.683	0.923	0.767
MCLEA	COLING'22	0.715	0.923	0.788	0.715	0.909	0.785	0.711	0.909	0.782
MSNEA	WWW'23	0.601	0.83	0.684	0.535	0.775	0.617	0.543	0.801	0.630
MEAformer	MM'23	0.771	0.951	0.835	0.764	0.959	0.834	0.770	0.961	0.841
PSNEA	MM'23	0.816	0.957	0.869	0.819	0.963	0.853	0.844	0.982	0.891
DESAlign	ICDE'24	0.868	0.969	0.909	0.871	0.974	0.913	0.882	0.982	0.924
IBMEA	MM'24	0.859	0.975	0.903	0.856	0.978	0.902	0.864	0.985	0.911
LoginMEA ^a	ECAI'24	0.877	0.972	0.905	0.868	0.972	0.910	0.879	0.983	0.919
OTMEA	ICASSP'25	0.779	0.953	0.842	0.783	0.959	0.849	0.796	0.97	0.861
AMFSEA ^a	COLING'25	0.691	0.879	0.751	0.696	0.871	0.757	0.767	0.914	0.818
ME3A-N	IPM'25	0.894	0.981	0.916	0.878	0.962	0.927	0.893	0.994	0.935
ME3A-M	IPM'25	0.886	0.984	0.919	0.882	0.970	0.929	0.895	0.994	0.938
HSP-N	–	0.897	0.987	0.924	0.894	0.982	0.936	0.903	0.993	0.944
HSP-M	–	0.902	0.986	0.930	0.889	0.985	0.938	0.906	0.995	0.946
$\Delta Imp.$	–	0.005	0.001	0.006	0.005	0.003	0.002	0.003	0.001	0.002
$\Delta Std.$	–	±0.004	±0.003	±0.005	±0.004	±0.003	±0.002	±0.003	±0.002	±0.004

^a Represents reproduced experimental results.

Table 4

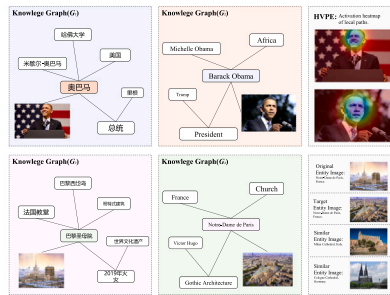
The experiment results of ablation study. The $\downarrow$ represents the difference between the current result and the model's final performance.

Component	DBP _{ZH-EN}			DBP _{JA-EN}		
	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
HSP (full)	0.902	0.986	0.930	0.889	0.985	0.938
HSP w/o HVPE	0.864 ($\downarrow$ 0.038)	0.964 ($\downarrow$ 0.022)	0.911 ($\downarrow$ 0.019)	0.852 ($\downarrow$ 0.037)	0.966 ($\downarrow$ 0.019)	0.920 ($\downarrow$ 0.018)
HSP w/o CSCN	0.870 ($\downarrow$ 0.032)	0.969 ($\downarrow$ 0.017)	0.916 ($\downarrow$ 0.014)	0.857 ($\downarrow$ 0.032)	0.970 ($\downarrow$ 0.015)	0.924 ($\downarrow$ 0.014)
HSP w/o EAFC	0.879 ($\downarrow$ 0.023)	0.977 ($\downarrow$ 0.009)	0.922 ($\downarrow$ 0.008)	0.868 ($\downarrow$ 0.021)	0.975 ($\downarrow$ 0.010)	0.929 ($\downarrow$ 0.009)
HSP w/o META	0.891 ($\downarrow$ 0.011)	0.982 ($\downarrow$ 0.004)	0.926 ($\downarrow$ 0.004)	0.879 ($\downarrow$ 0.010)	0.981 ($\downarrow$ 0.004)	0.934 ($\downarrow$ 0.004)
HSP w/o CONT	0.874 ($\downarrow$ 0.028)	0.972 ($\downarrow$ 0.014)	0.918 ($\downarrow$ 0.012)	0.861 ($\downarrow$ 0.028)	0.969 ($\downarrow$ 0.016)	0.923 ($\downarrow$ 0.015)

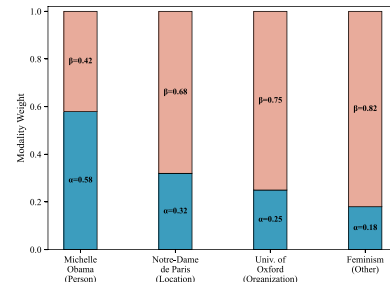
Table 5

Efficiency and performance analysis of different visual and text encoders.

Vis. pre-trained	Txt. pre-trained	Vis. Param.	Txt. Param.	Infer time (ms)	FB15K-DB15K		
					Hits@1	Hits@10	MRR
ResNet-50	BERT-base	26M	110M	27.8	0.746	0.883	0.785
ResNet-152	BERT-base	60M	110M	35.2	0.755	0.894	0.807
VGG-16	BERT-base	138M	110M	38.1	0.751	0.888	0.801
ViT-base	BERT-base	87M	110M	39.6	0.753	0.891	0.804
CLIP-ViT-large	BERT-base	303M	110M	78.3	0.760	0.900	0.813
ResNet-152	Glove-300d	60M	(static)	23.9	0.594	0.730	0.615
ResNet-152	Albert-base	60M	11M	26.4	0.692	0.825	0.728
ResNet-152	BERT-base	60M	110M	35.2	0.755	0.894	0.807
ResNet-152	BERT-large	60M	340M	52.1	0.758	0.899	0.811



(a) Sample entity.



(b) Modality weights of different entity type.

Fig. 3. Case study.

cost. The full HSP model contains about 145M parameters with an inference latency of 40.0 ms, indicating that its computational scale remains within a practical deployment range for large-scale alignment.

As illustrated in Fig. 4, we train a lightweight single-path student network via knowledge distillation to reduce computational burden. The distilled model reduces latency to 31.4 ms, achieving a 21.5%

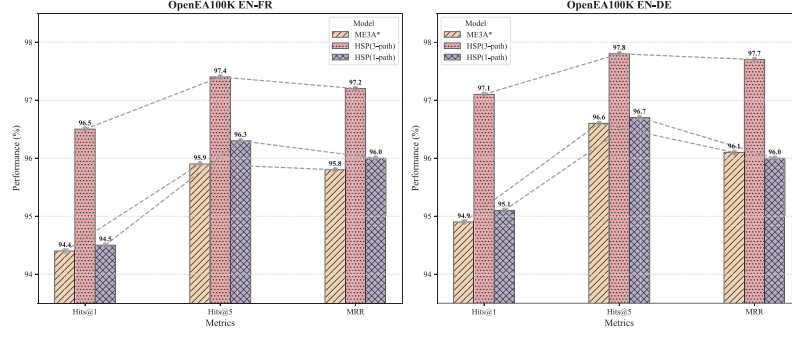
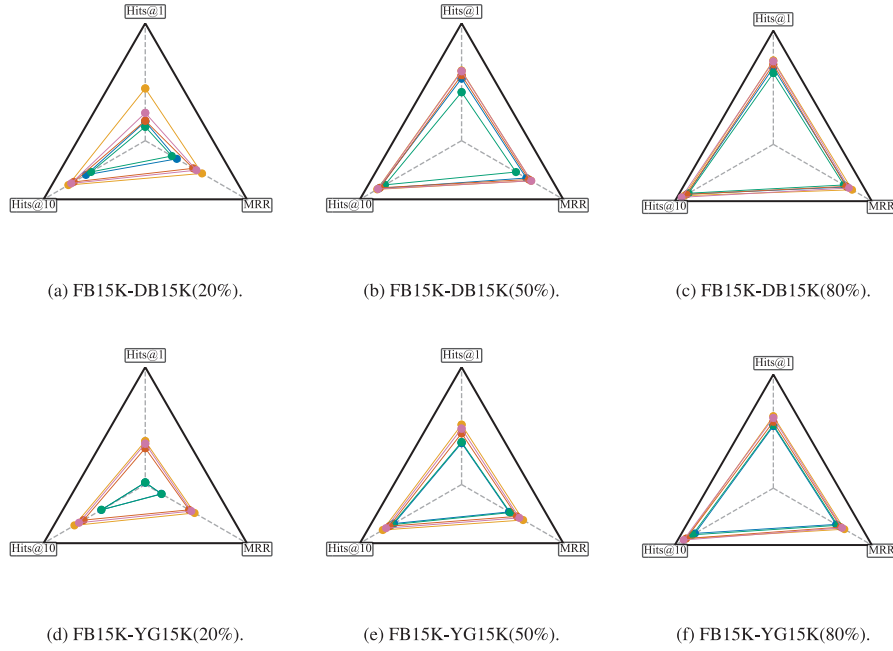


Fig. 4. The experiment results of efficiency analysis.

Fig. 5. Performance comparison of baseline models. The $\bullet$, $\circ$, $\triangle$, $\square$, and $\star$ represent DESAlign, LoginMEA, RICEA, ME3A-M, and HSP-M models, respectively.

speedup with only a 2.0% drop in Hits@1. This lightweight variant outperforms ME3A on the EN-FR and EN-DE pairs, indicating that HSP maintains a favorable balance between alignment accuracy and computational efficiency.

We further conduct a comparative analysis of HSP against representative methods to evaluate deployment efficiency. HSP (3-Path) exhibits slightly higher latency than ME3A but remains substantially faster than CDMEA and PCMEA. CDMEA and PCMEA rely on causal graph decomposition with counterfactual interventions and pseudo-label driven iterative enhancement, respectively, incurring higher inference overhead. HSP leverages structured multi-path feature fusion to maintain high alignment accuracy with computational cost within a practical range. The distilled variant achieves lower inference latency and continues to outperform existing competitive approaches, indicating that HSP is practical and scalable for large-scale knowledge graph alignment.

4.9. Reproducibility analysis

In the main experiments, the proposed HSP consistently outperforms a range of recent competitive methods. However, we observe that several models exhibit performance fluctuations across different runs, indicating sensitivity to random initialization and training stochasticity,

as shown in Fig. 5. To systematically assess model stability, we select multiple representative approaches and conduct repeated independent experiments on two cross-graph and two cross-lingual datasets. To reduce the influence of supervision strength on performance variance, all experiments are repeated five times under a moderate supervision setting.

As illustrated in Fig. 6, HSP demonstrates strong stability across multiple runs, with the variance of all evaluation metrics remaining below 1.5% and performance distributions highly concentrated. In contrast, although LoginMEA achieves relatively high peak performance, it exhibits more noticeable fluctuations, with a variance of approximately 1.8%. Furthermore, DESAlign, RICEA, and ME³A show variances exceeding 1.4% across datasets, suggesting greater sensitivity to randomness. Overall, while maintaining favorable performance, HSP exhibits improved stability and consistency, resulting in smaller performance variations and more reliable generalization across both cross-lingual and cross-graph scenarios.

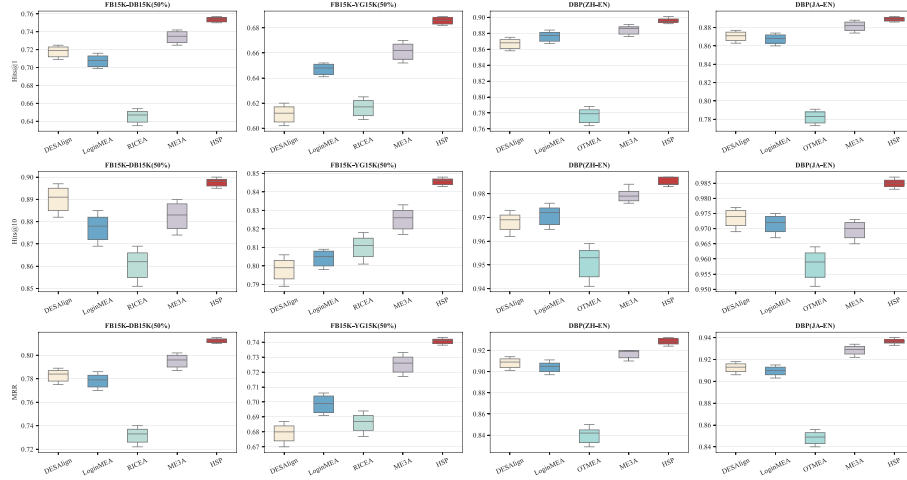
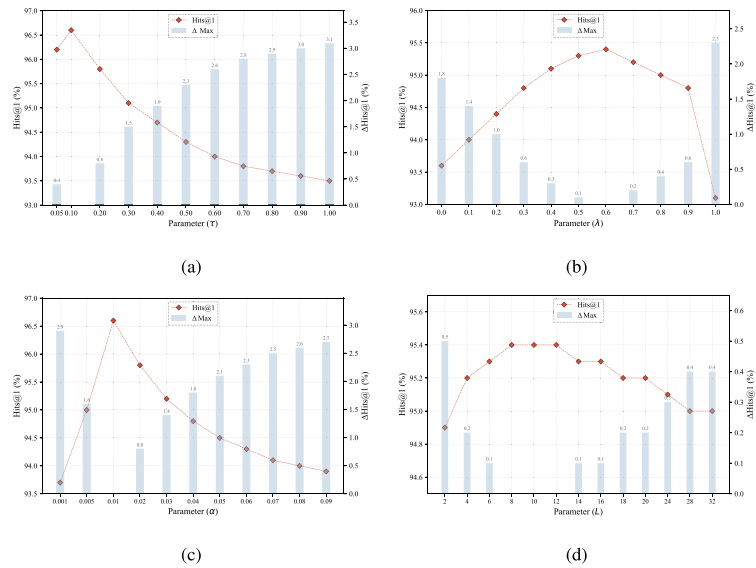
4.10. Hyperparameters sensitivity analysis

We conduct a systematic sensitivity analysis of four key hyperparameters on the OpenEA100K (EN-FR) dataset to assess the robustness of HSP, as its larger scale and more diverse entity distribution provide

Table 6

Model parameter counts and per-sample inference time on the OpenEA100K dataset. T.C. denotes time cost, and Param. denotes the number of parameters. Inference latency is reported as the average of three runs under identical hardware settings.

Component	Param.	T.C. (ms)	Core computation cost analysis
HVPE	25.6M	–	Parallel path of three-layer Transformer.
CSCN	8.1M	–	Cross modal projection and contrastive loss calculation.
EAFC	1.2M	–	Gated Networks and Meta Learning (Training).
PLM (BERT)	110M	–	Standard BERT encoding and implicit reasoning.
HSP (3-Path)	145M	40	–
HSP (1-Path)	130M	31.4	–
ME ³ A	–	35.2	Mapping all modalities to text sequences with PLM-based entailment inference.
PCMEA	–	96.3	Dynamic pseudo-labeling with confidence calibration and iterative refinement.
CDMEA	–	42.7	Causal graph decomposition with counterfactual construction and intervention.
MFIEA	–	21.5	Feature-level interaction with non-sequential modeling; conventional architecture.
IBMEA	–	28.4	Information bottleneck compression with variational inference and MI estimation.

**Fig. 6.** Performance distribution of the model over five independent runs on the same platform.**Fig. 7.** The performance changes of the model under different parameter settings.

a more statistically reliable testbed than DBP15K. As shown in Fig. 7, the model exhibits stable performance across a wide range of settings. The contrastive loss temperature τ achieves peak Hits@1 (95.4%) at $\tau = 0.1$; smaller or larger values degrade performance due to overfitting or insufficient discrimination. The fusion balance coefficient λ

is optimal at 0.6, confirming the necessity of combining embedding retrieval with semantic re-ranking. The EAFC inner-loop learning rate $\alpha = 0.01$ balances fast adaptation and generalization, while visual prefix length $L = 8$ offers the best trade-off between accuracy and efficiency, with performance remaining above 95.2% for $L \in [4, 16]$.

These results demonstrate that HSP is not only effective but also robust to hyperparameter variations, supporting its practical applicability.

5. Limitations analysis

Although HSP achieves superior performance, it has three practical limitations. First, the HSP model relies on frozen ResNet-152 features as visual input. Removing HVPE reduces Hits@1 by 3.8% on DBP_{ZH-EN}, showing dependence on high-quality visual features; however, ResNet-152 may provide insufficient representations for abstract or visually ambiguous entities (e.g., “inflation”), limiting alignment. This limitation could be mitigated by adopting stronger vision foundation models. Second, the EAFC component assumes a predefined set of entity types for adaptive fusion. Although meta-learning aids generalization, alignment of unseen or fine-grained types depends on an auxiliary type predictor, which may propagate errors into fusion weights, restricting applicability in open-world settings. Future work could explore type-free fusion mechanisms that adapt without explicit type supervision. Third, the model-agnostic meta-learning in EAFC introduces non-negligible training overhead due to bi-level optimization, which may limit scalability on ultra-large knowledge graphs under memory-constrained batch processing. Future work could investigate first-order approximations or distributed strategies to alleviate this cost.

6. Conclusion

In this paper, we re-examine entailment-based multimodal entity alignment and identify three fundamental limitations that hinder its effectiveness: (1) semantic collapse caused by aggressive compression of high-dimensional visual features, (2) insufficient cross-modal semantic consistency due to weak calibration mechanisms, and (3) rigid fusion strategies that ignore entity-type-specific modality relevance. To address these challenges, we propose HSP, a Hierarchical Semantic-Preserving framework that unifies the alignment process under a Preserve–Calibrate–Adapt paradigm. Concretely, HSP preserves the intrinsic topology and fine-grained semantics of visual representations via a hierarchical vision-preserving encoder, thereby mitigating information loss during feature compression. It then enforces cross-modal semantic consistency by explicitly calibrating visual and textual representations within a shared semantic space. Furthermore, HSP introduces an entity-type-adaptive fusion controller that dynamically adjusts modality contributions; the fusion strategy is optimized via meta-learning to capture context-dependent modality relevance. By jointly addressing semantic fidelity, cross-modal consistency, and adaptive fusion, HSP improves the reliability and expressiveness of entailment-based alignment. Extensive experiments on five public benchmarks across cross-lingual and cross-graph scenarios show that it consistently outperforms competitive baselines under varying supervision levels, demonstrating strong effectiveness and generalization.

CRedit authorship contribution statement

Bao Wang: Writing – original draft. **Ruizhe Ma:** Writing – review & editing. **Xinyu Liang:** Validation, Data curation. **Li Yan:** Validation. **Zongmin Ma:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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